

Multiplication law



1

a y^5

b $8x^9$

c $12y^7$

d x^{y+z}

e $-6a^{12}$

f $-10a^5$

g b^{13}

h a^{b+c}

i $30y^8$

j $24u^4$

k $20b^{13}$

l x^{a+b+c}

2

a 3

b $5x^{12}$

c -2

d $6x^5$

Division law

3

a a^4

b $\frac{x^3}{5}$

c $6m^{11}$

d b^4

e j^7

f x^3

g f^3

h s^9

i $2k^5$

j $8g^7$

k $3p^3$

l $5l^6$

4

a 4

b 5

c 11

d 4

Power of a power law

5

a

i Yes

ii Yes

iii Yes

iv No

v No

b

i False

ii True

c 6

6

a j^{18}

b w^8

c t^{12}

d $27a^{12}$

e $125h^{21}$

f x^5y^{10}

g $\frac{16}{h^8}$

h $\frac{a^6}{b^{15}}$

i $\frac{27x^6}{y^{15}}$

j x^{36}

k $-125x^{12}$

l $-\frac{8a^{12}}{27b^6}$



Zero index

8

a 18

e 9

b 1

f 1

c 1

g $\frac{3}{t^2}$

d 1

h 8

9

a 0

b 8

c 9

d 0

Mixed laws

10

a $x^{24}y^{12}$

e m^8

i 0

m $16u^{10}v^4$

q $16x^3$

u 3^{a+6}

b y^8

f p^2

j -10

n $81a^8b^{20}c^4$

r c^{86}

v 3^{43a-21}

c p^{17}

g $16x^{12}$

k 22

o $1125y^{16}$

s $\frac{3a^2}{b}$

d $8y^6$

h $3p^9$

l x^{60}

p x^3

t -9

11 2^{16p}